A FUNCTORIAL INJECTION FROM EXTENSION BY ZERO INTO THE DIRECT IMAGE FOR A SEPARATED ETALE MORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the abelian case is written out and the module case is covered by the same construction.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Extension by zero j_! and direct image j_* along a separated etale morphism. Same chapter, the functors j_! and j_* attached to j : U -> X and their description by sections and by compactly supported sections. Supplies the two functors and the map compared in the statement.
- Separation of j and the description of sections of the direct image. Same chapter, the consequences of separatedness of j for the description of sections of j_* and for the finiteness of the supports appearing in the colimits defining j_!. Supplies the injectivity argument, in which a finite sum of sections with disjoint supports is examined.

Source reference. Original equation 0F4K: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11405–11405. Navigation pointer invoked at proof line 11530; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 01KT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11545–11545. Navigation pointer invoked at proof line 11545; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $j : U \to X$ be separated and étale. Then there is a functorial injective map $j_!\mathcal{F} \to j_*\mathcal{F}$ on abelian sheaves and sheaves of Λ-modules.*

Proof. We prove this in the case of abelian sheaves. Let us construct a canonical map

$$j_{p!}\mathcal{F} \to j_*\mathcal{F}$$

of abelian presheaves on $X_{étale}$ for any abelian sheaf $\mathcal{F}$ on $U_{étale}$ where $j_{p!}$ is as in (0F4K). Sheafification of this map will be the desired map $j_!\mathcal{F} \to j_*\mathcal{F}$. Evaluating both sides on $V \to X$ étale we obtain

$$j_{p!}\mathcal{F}(V) = \bigoplus\nolimits_{\varphi : V \to U} \mathcal{F}(V \xrightarrow{\varphi} U) \quad \text{and} \quad j_*\mathcal{F}(V) = \mathcal{F}(V \times_X U)$$

For each φ we have an open and closed immersion

$$\Gamma_\varphi = (1, \varphi) : V \longrightarrow V \times_X U$$

over U. It is open as it is a morphism between schemes étale over U and it is closed as it is a section of a scheme separated over V (Schemes, Lemma 01KT). Thus for a section $s_\varphi \in \mathcal{F}(V \xrightarrow{\varphi} U)$ there exists a unique section s'_φ in $\mathcal{F}(V \times_X U)$ which pulls back to s_φ by Γ_φ and which restricts to zero on the complement of the image of Γ_φ.

To show that our map is injective suppose that $\sum_{i=1,\ldots,n} s_{\varphi_i}$ is an element of $j_{p!}\mathcal{F}(V)$ in the formula above maps to zero in $j_*\mathcal{F}(V)$. Our task is to show that $\sum_{i=1,\ldots,n} s_{\varphi_i}$ restricts to zero on the members of an étale covering of V. Looking at all pairwise equalizers (which are open and closed in V) of the morphisms $\varphi_i : V \to U$ and working locally on V, we may assume the images of the morphisms $\Gamma_{\varphi_1}, \ldots, \Gamma_{\varphi_n}$ are pairwise disjoint. Since our assumption is that $\sum_{i=1,\ldots,n} s'_{\varphi_i} = 0$ we then immediately conclude that $s'_{\varphi_i} = 0$ for each i (by the disjointness of the supports of these sections), whence $s_{\varphi_i} = 0$ for all i as desired. □